

Meghana Ganapa

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Summary

Analytics and Data Engineering graduate with hands-on experience building cloud-based ETL pipelines, AI-powered automation systems, and scalable data workflows using Python, SQL, AWS, and LLM technologies. Skilled in data modelling, workflow automation, data validation, CI/CD concepts, and backend API development, with a strong interest in healthcare analytics and AI-driven decision-making.

Education

The University of Melbourne

Feb 2023 - Dec 2024

Masters, Data Science

- GPA: 7.8

Relevant Experience

Melbourne Data Analytics Platform

Feb 2023 - May 2023

Research Intern

Melbourne, Victoria

- Conducted comparative analysis of ImageJ and Pixel Annotation Tool to support development of a CNN-based image classification model for gonadal tissue analysis. Implemented CNN in Python, improving tissue classification accuracy by 15%.

Qentelli Solutions Private Limited

Sep 2022 - Dec 2022

Associate Software Engineer

Hyderabad, Telangana

- Optimized SQL queries and automated REST API workflows, reducing manual processing by 30% and improving data retrieval efficiency for production systems.
- Collaborated with cross-functional teams to translate requirements into data-driven automation and API-based solutions.

Accolite Digital

Feb 2022 - Aug 2022

Software Engineer Intern

Hyderabad, Telangana

- Developed and deployed a robust Model-View-Controller (MVC) architecture using Spring Boot and MongoDB. This resulted in a 30% increase in development velocity, a 25% improvement in system reliability, and a significant reduction in production error rates.

Projects

AI Coding Agent using Codex and LLM Workflows

May 2026

- Python, OpenAI Codex, FastAPI, REST APIs, LLMs, Prompt Engineering
- Designed and developed an AI-powered coding assistant agent using OpenAI Codex and LLM orchestration workflows to automate code generation, debugging, and developer productivity tasks.
- Built scalable backend APIs using FastAPI to support prompt processing, context management, and AI-assisted workflow automation.
- Implemented prompt engineering, API integration, and response validation techniques to improve code quality, accuracy, and workflow efficiency.
- Applied software engineering best practices including modular architecture, backend automation, and production-focused AI system design.

Retrieval-Augmented Generation (RAG) System for Medical Textbooks

Apr 2025 - Jun 2025

- Python, SentenceTransformers, Vector Databases, CrossEncoder, LLMs
- Engineered a Retrieval-Augmented Generation (RAG) pipeline to enable AI-powered question answering over large-scale medical textbooks using semantic search and large language models.
- Implemented document chunking, embedding generation, vector similarity search, and CrossEncoder reranking pipelines to improve retrieval accuracy and response relevance.
- Built scalable AI retrieval workflows using vector databases and embedding-based search, reducing manual document search time by approximately 40%.
- Applied prompt engineering, retrieval optimisation, and LLM evaluation techniques to improve response quality and contextual relevance.

Roo Jobs Daily – AI and Startup Jobs Aggregation Pipeline

Apr 2026

- Python, FastAPI, PostgreSQL, Slack API, Playwright, REST APIs, Workflow Automation
- Designed and developed an automated daily jobs pipeline to aggregate, filter, rank, and deliver AI and startup job opportunities relevant to Australia and remote roles.
- Built backend workflows using Python and FastAPI to collect fresh job listings from multiple public job sources, perform deduplication, and categorise opportunities into Australian AI, Australian Startup, Remote AI, and Remote Startup roles.
- Implemented automated scheduling and Slack integration to publish the top 7 curated job opportunities into Slack channels every day at 7:00am Australia/Melbourne time, alongside links to a full list of matched jobs.
- Developed ranking and filtering logic to improve job relevance, reduce noisy results, and ensure reliable workflow execution with logging, monitoring, and failure-safe alerting mechanisms.

- Python, NLP, BERT, BiLSTM, TensorFlow, PyTorch, Flask
- Developed an NLP and machine learning pipeline using BERT and BiLSTM models to extract clinically relevant information from unstructured pathology reports.
- Built predictive models to identify polyp size and histology from clinical text, enabling automated recommendation support for colonoscopy follow-up decisions.
- Engineered custom text parsing and feature extraction algorithms for complex clinical dimensional expressions, achieving approximately 85% classification accuracy.
- Deployed Flask-based APIs to convert unstructured medical reports into structured healthcare datasets, reducing manual clinical review effort by approximately 30%.

Technical Skills

- **Programming Languages:** Python, SQL, R, Java, TypeScript
- **Analytics Engineering & Data Engineering:** ETL/ELT Pipelines, Data Modelling (Star Schema), dbt Concepts, Data Validation, Data Governance, Data Quality Automation, Workflow Optimisation, CI/CD Concepts
- **Cloud & Platforms:** AWS (S3, Glue, Lambda, Redshift, EC2), Azure, GitHub Actions, Docker (basic)
- **AI & Machine Learning:** Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs)-, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Automation, Predictive Modelling
- **Business Intelligence & Analytics:** Power BI (DAX, Power Query), Tableau, Dashboard Development, Data Analysis, Reporting, Semantic Data Modelling
- **Backend & APIs:** Flask, FastAPI, REST APIs, API Integration, Backend Workflow Automation
- **Databases:** PostgreSQL, MySQL, MongoDB, Neo4j, Vector Databases (Qdrant), Relational Databases
- **Tools & Version Control:** Git, GitHub, Jupyter Notebook, VS Code
- **Soft Skills:** Stakeholder Communication, Problem Solving, Cross-functional Collaboration, Technical Documentation

Achievements

- Published two peer-reviewed research papers in Springer on machine learning applications in healthcare analytics: "Cognitive Baby Care Solutions for Smart Parenting" and "Effective Prediction of Brain Tumor Using Machine Learning Algorithms".
- Participated in Smart India Hackathon-2018, I-Innovate, and IBM hackathons, showcasing innovative solutions to technical challenges.

Certifications

- AWS Cloud Practitioner